Exp 9

Aim - To implement a Private Ethereum Blockchain using Geth.

Theory -

Ethereum is an open-source, decentralized blockchain platform that enables developers to build and deploy smart contracts and decentralized applications (DApps). Unlike public blockchains such as the Ethereum mainnet, a Private Ethereum Blockchain is restricted to specific participants, making it ideal for development, testing, and enterprise applications where confidentiality, control, and performance are essential.

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through JSON-RPC, command-line, or JavaScript consoles.

Setting up a Private Ethereum Network involves creating a customized blockchain environment isolated from the public Ethereum network. It enables developers to test smart contracts, consensus mechanisms, and network behavior without incurring gas costs or depending on public validators.

Steps Involved:

1. **Choosing a Network ID:**
2. Each private blockchain must have a unique network ID to differentiate it from public or other private networks.
3. **Choosing a Consensus Algorithm:**
4. Ethereum supports different consensus mechanisms such as **Proof of Work (PoW)** and **Proof of Authority (PoA)**. In private networks, **Clique (PoA)** is often used for faster block confirmation.
5. **Creating a Genesis Block:**
6. The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.
7. **Initializing the Geth Database:**
8. The `geth init` command initializes the blockchain using the **genesis.json** file, preparing the data directory for node operation.
9. **Setting up Networking:**
10. Nodes communicate using **enodes** (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.
11. **Running the Member Nodes:**

12. Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.
13. **Running a Signer (in Clique):**
14. In the **Clique consensus**, designated **signers** validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

Outputs -

1. First install all the necessary dependencies like curl, git, build-essential.
   sudo apt install wget curl git build-essential -y

```
deilsy@delsy:~$ sudo apt install wget curl git build-essential -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
git is already the newest version (1:2.43.0-1ubuntu7.3).
build-essential is already the newest version (12.10ubuntu1).
The following additional packages will be installed:
  libcurl3t64-gnutls libcurl4t64
The following packages will be upgraded:
  curl libcurl3t64-gnutls libcurl4t64
3 upgraded, 0 newly installed, 0 to remove and 57 not upgraded.
Need to get 902 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8
.5.0-2ubuntu10.8 [227 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl4t64
amd64 8.5.0-2ubuntu10.8 [342 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-
gnutls amd64 8.5.0-2ubuntu10.8 [334 kB]
Fetched 902 kB in 1s (650 kB/s)
(Reading database ... 46694 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.8_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.8) over (8.5.0-2ubuntu10.6) ...
Preparing to unpack .../libcurl4t64_8.5.0-2ubuntu10.8_amd64.deb ...
Unpacking libcurl4t64:amd64 (8.5.0-2ubuntu10.8) over (8.5.0-2ubuntu10.6) ...
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu10.8_amd64.deb ...
Unpacking libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.8) over (8.5.0-2ubuntu10
.6) ...
Setting up libcurl4t64:amd64 (8.5.0-2ubuntu10.8) ...
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.8) ...
Setting up curl (8.5.0-2ubuntu10.8) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
```

2. Install geth using wget cd /usr/local/bin && sudo wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0- e5 01b3b0.tar.gz

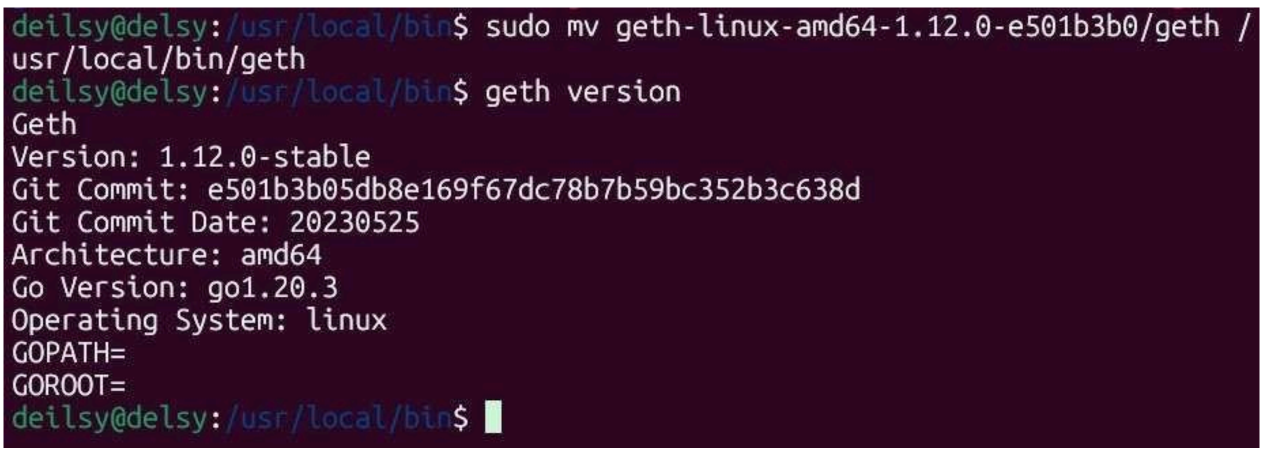
```
deilsy@delsy:~$ cd /usr/local/bin && sudo wget https://gethstore.blob.core.w
indows.net/builds/geth-linux-amd64-1.12.0-e501b3b0.tar.gz
--2026-03-26 03:54:03--  https://gethstore.blob.core.windows.net/builds/geth
-linux-amd64-1.12.0-e501b3b0.tar.gz
Resolving gethstore.blob.core.windows.net (gethstore.blob.core.windows.net).
.. 20.60.40.164
Connecting to gethstore.blob.core.windows.net (gethstore.blob.core.windows.n
et)|20.60.40.164|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 26856444 (26M) [application/octet-stream]
Saving to: ‘geth-linux-amd64-1.12.0-e501b3b0.tar.gz’

geth-linux-amd64-1 100%[================>]  25.61M   734KB/s    in 27s

2026-03-26 03:54:32 (968 KB/s) - ‘geth-linux-amd64-1.12.0-e501b3b0.tar.gz’ s
aved [26856444/26856444]

deilsy@delsy:/usr/local/bin$ ls
geth-linux-amd64-1.12.0-e501b3b0.tar.gz
deilsy@delsy:/usr/local/bin$
```

3. Extract the tar file and move the geth into /usr/local/bin/geth path & check the geth version
   sudo tar -xvf geth-linux-amd64-1.12.0-e501b3b0.tar.gz sudo mv
   geth-linux-amd64-1.12.0-e501b3b0/geth /usr/local/bin/geth geth version



```
deilsy@delsy:/usr/local/bin$ sudo mv geth-linux-amd64-1.12.0-e501b3b0/geth /
usr/local/bin/geth
deilsy@delsy:/usr/local/bin$ geth version
Geth
Version: 1.12.0-stable
Git Commit: e501b3b05db8e169f67dc78b7b59bc352b3c638d
Git Commit Date: 20230525
Architecture: amd64
Go Version: go1.20.3
Operating System: linux
GOPATH=
GOROOT=
deilsy@delsy:/usr/local/bin$
```

4. Create a folder named private_ethereum_setup and create another 2 directories naming node1 and node2 individually inside private_ethereum_setup directory.
   mkdir private_ethereum_setup
   cd private_ethereum_setup && mkdir node1 node2

```
deilsy@delsy:~$ ls private_ethereum_setup/
node1  node2
deilsy@delsy:~$
```

5. Check the files
6. Create account for node1
   geth --datadir "node1" account new

```
deilsy@delsy:~/private_ethereum_setup$ geth --datadir "node1" account new
INFO [03-26|04:01:54.815] Maximum peer count                       ETH=50 LE
S=0 total=50
INFO [03-26|04:01:54.816] Smartcard socket not found, disabling    err="stat
 /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not f
orget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x563A5a670260417B163AD1696fCBfA6849906d06
Path of the secret key file: node1/keystore/UTC--2026-03-26T04-02-01.7653580
44Z--563a5a670260417b163ad1696fcbfa6849906d06

- You can share your public address with anyone. Others need it to interact
with you.
- You must NEVER share the secret key with anyone! The key controls access t
o your funds!
- You must BACKUP your key file! Without the key, it's impossible to access
account funds!
- You must REMEMBER your password! Without the password, it's impossible to
decrypt the key!

deilsy@delsy:~/private_ethereum_setup$
```

7. Do the same for node2
   geth --datadir "node2" account new

```
deilsy@delsy:~/private_ethereum_setup$ geth --datadir "node2" account new
INFO [03-26|04:02:43.813] Maximum peer count                       ETH=50 LE
S=0 total=50
INFO [03-26|04:02:43.813] Smartcard socket not found, disabling    err="stat
 /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not f
orget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x6d04eA66725A40F5669E2d6f4f77eb4Ecf756a3f
Path of the secret key file: node2/keystore/UTC--2026-03-26T04-02-49.2597870
89Z--6d04ea66725a40f5669e2d6f4f77eb4ecf756a3f

- You can share your public address with anyone. Others need it to interact
with you.
- You must NEVER share the secret key with anyone! The key controls access t
o your funds!
- You must BACKUP your key file! Without the key, it's impossible to access
account funds!
- You must REMEMBER your password! Without the password, it's impossible to
decrypt the key!
```

8. Create a file called genesis.json in the same directory and copy paste these contents

```
{
   "config": { "chainId": 12345,
      "homesteadBlock": 0,
      "eip150Block": 0,
      "eip155Block": 0,
      "eip158Block": 0,
      "byzantiumBlock": 0,
```

```
        "constantinopleBlock": 0,
        "petersburgBlock": 0,
        "istanbulBlock": 0,
        "muirGlacierBlock": 0,
        "berlinBlock": 0,
        "londonBlock": 0,
        "arrowGlacierBlock": 0,
        "grayGlacierBlock": 0, "clique": {
            "period": 5,
            "epoch": 30000
        }
    },
    "difficulty": "1",
    "gasLimit": "800000000", "extradata":
"0x00000000000000000000000000000000000000000000000000000000
0000000098608ADf9c785d54f40cDcf6700E990771b192260000000000000
0000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000
0",
    "alloc": { "0x98608ADf9c785d54f40cDcf6700E990771b19226": {
"balance": "10000000000000000000000" },
        "0x7B25e791D24A3F5c453A9E5468cF6cEa2243092C": {
"balance": "12000000000000000000000" }
    }
}
```

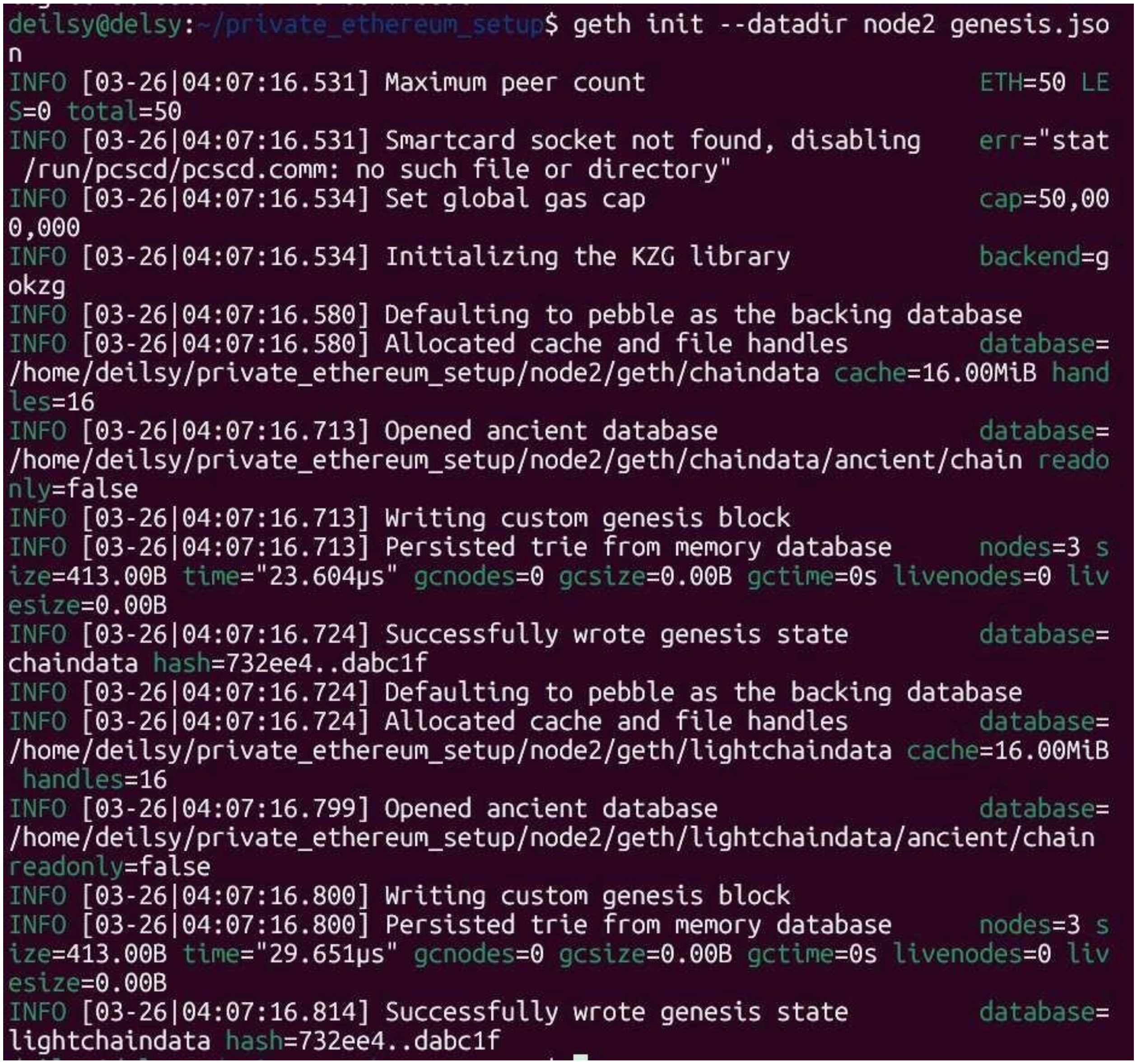
Make changes in the alloc part replace the public keys of the account that you have generated in earlier steps.

```
{
  "config": {
    "chainId": 12345,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "istanbulBlock": 0,
    "muirGlacierBlock": 0,
    "berlinBlock": 0,
    "londonBlock": 0,
    "arrowGlacierBlock": 0,
    "grayGlacierBlock": 0,
    "clique": {
      "period": 5,
      "epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "800000000",
  "extradata": "0x000000000000000000000000000000000000000000000000000000000>
  "alloc": {
    "0x563A5a670260417B163AD1696fCBfA6849906d06": { "balance": "10000000000>
    "0x6d04eA66725A40F5669E2d6f4f77eb4Ecf756a3f": { "balance": "12000000000>
  }
}
```

9. Initializing geth for node1 using the genesis.json file
   geth init --datadir node1 genesis.json

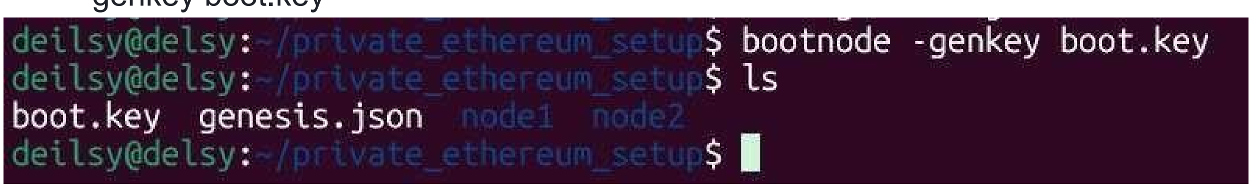
```
deilsy@delsy:~/private_ethereum_setup$ geth init --datadir node1 genesis.json
INFO [03-26|04:06:53.763] Maximum peer count                       ETH=50 LES=0 total=50
INFO [03-26|04:06:53.764] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [03-26|04:06:53.767] Set global gas cap                       cap=50,000,000
INFO [03-26|04:06:53.768] Initializing the KZG library             backend=gokzg
INFO [03-26|04:06:53.810] Defaulting to pebble as the backing database
INFO [03-26|04:06:53.810] Allocated cache and file handles         database=/home/deilsy/private_ethereum_setup/node1/geth/chaindata cache=16.00MiB handles=16
INFO [03-26|04:06:53.957] Opened ancient database                  database=/home/deilsy/private_ethereum_setup/node1/geth/chaindata/ancient/chain readonly=false
INFO [03-26|04:06:53.957] Writing custom genesis block
INFO [03-26|04:06:53.958] Persisted trie from memory database      nodes=3 size=413.00B time="100.853µs" gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [03-26|04:06:53.976] Successfully wrote genesis state         database=chaindata hash=732ee4..dabc1f
INFO [03-26|04:06:53.976] Defaulting to pebble as the backing database
INFO [03-26|04:06:53.976] Allocated cache and file handles         database=/home/deilsy/private_ethereum_setup/node1/geth/lightchaindata cache=16.00MiB handles=16
INFO [03-26|04:06:54.074] Opened ancient database                  database=/home/deilsy/private_ethereum_setup/node1/geth/lightchaindata/ancient/chain readonly=false
INFO [03-26|04:06:54.074] Writing custom genesis block
INFO [03-26|04:06:54.074] Persisted trie from memory database      nodes=3 size=413.00B time="18.276µs"  gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [03-26|04:06:54.086] Successfully wrote genesis state         database=lightchaindata hash=732ee4..dabc1f
deilsy@delsy:~/private_ethereum_setup$
```

10. Do the same for node2
    geth init --datadir node2 genesis.json



```
deilsy@delsy:~/private_ethereum_setup$ geth init --datadir node2 genesis.jso
n
INFO [03-26|04:07:16.531] Maximum peer count                       ETH=50 LE
S=0 total=50
INFO [03-26|04:07:16.531] Smartcard socket not found, disabling    err="stat
 /run/pcscd/pcscd.comm: no such file or directory"
INFO [03-26|04:07:16.534] Set global gas cap                       cap=50,00
0,000
INFO [03-26|04:07:16.534] Initializing the KZG library             backend=g
okzg
INFO [03-26|04:07:16.580] Defaulting to pebble as the backing database
INFO [03-26|04:07:16.580] Allocated cache and file handles         database=
/home/deilsy/private_ethereum_setup/node2/geth/chaindata cache=16.00MiB hand
les=16
INFO [03-26|04:07:16.713] Opened ancient database                  database=
/home/deilsy/private_ethereum_setup/node2/geth/chaindata/ancient/chain reado
nly=false
INFO [03-26|04:07:16.713] Writing custom genesis block
INFO [03-26|04:07:16.713] Persisted trie from memory database      nodes=3 s
ize=413.00B time="23.604µs" gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 liv
esize=0.00B
INFO [03-26|04:07:16.724] Successfully wrote genesis state         database=
chaindata hash=732ee4..dabc1f
INFO [03-26|04:07:16.724] Defaulting to pebble as the backing database
INFO [03-26|04:07:16.724] Allocated cache and file handles         database=
/home/deilsy/private_ethereum_setup/node2/geth/lightchaindata cache=16.00MiB
 handles=16
INFO [03-26|04:07:16.799] Opened ancient database                  database=
/home/deilsy/private_ethereum_setup/node2/geth/lightchaindata/ancient/chain
readonly=false
INFO [03-26|04:07:16.800] Writing custom genesis block
INFO [03-26|04:07:16.800] Persisted trie from memory database      nodes=3 s
ize=413.00B time="29.651µs" gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 liv
esize=0.00B
INFO [03-26|04:07:16.814] Successfully wrote genesis state         database=
lightchaindata hash=732ee4..dabc1f
```

11. Now we will install bootnode for configuring both the nodes. sudo add-apt-repository -y ppa:ethereum/ethereum sudo apt-get update sudo apt-get install bootnode
12. Create a key for the bootnode and save it to boot.key in the folder bootnode -genkey boot.key



```
deilsy@delsy:~/private_ethereum_setup$ bootnode -genkey boot.key
deilsy@delsy:~/private_ethereum_setup$ ls
boot.key  genesis.json  node1  node2
deilsy@delsy:~/private_ethereum_setup$
```

13. Copy this command and copy the enode://…. This will be required later bootnode -nodekey boot.key -verbosity 9 -addr :30305

```
deilsy@delsy:~/private_ethereum_setup$ bootnode -nodekey boot.key -verbosity
 9 -addr :30305
enode://3f690f88c00ce563d9dd609b0674ad97df9fc6b77673bd86917f51333118e81f54e7
5919746a331d55af7f6edba870a5ef5ed12b60af554ccb922bc77ab41f27@127.0.0.1:0?dis
cport=30305
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deploymen
ts.
INFO [03-26|04:20:52.969] New local node record                    seq=1,774
,498,852,968 id=7212962da899bfc0 ip="invalid IP" udp=0 tcp=0
```

14. Now on another terminal we will start mining node1

geth --datadir node1 --port 30306 --bootnodes enode://3f690f88c00ce563d9dd609b0674ad97df9fc6b77673bd86917f51333118e81f54e75919746a331d55af7f6edba870a5ef5ed12b60af554ccb922bc77ab41f27@127.0.0.1:0?discport=30305 --networkid 123454321 -- unlock 0x563A5a670260417B163AD1696fCBfA6849906d06 --password node1/password.txt --authrpc.port 8551 --miner.etherbase 0x563A5a670260417B163AD1696fCBfA6849906d06 --mine

Replace the address with your own created

```
deilsy@delsy:~/private_ethereum_setup$ geth --datadir node1 --port 30306 --bootnodes enode://3f690f88c00ce563d9dd609b0674ad97df9fc6b77673bd86917f51333118e81f54e75919746a331d55af7f6edba870a5ef5ed12b60af554ccb922bc77ab41f27@127.0.0.1:0?discport=30305 --networkid 123454321 --unlock 0x563A5a670260417B163AD1696fCBfA6849906d06 --password node1/password.txt --authrpc.port 8551 --miner.etherbase 0x563A5a670260417B163AD1696fCBfA6849906d06 --mine
INFO [03-26|04:28:57.471] Maximum peer count                       ETH=50 LES=0 total=50
INFO [03-26|04:28:57.472] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [03-26|04:28:57.474] Set global gas cap                       cap=50,000,000
INFO [03-26|04:28:57.474] Initializing the KZG library             backend=gokzg
INFO [03-26|04:28:57.529] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [03-26|04:28:57.529] Using pebble as the backing database
INFO [03-26|04:28:57.530] Allocated cache and file handles         database=/home/deilsy/private_ethereum_setup/node1/geth/chaindata cache=512.00MiB handles=524,288
INFO [03-26|04:28:57.629] Opened ancient database                  database=/home/deilsy/private_ethereum_setup/node1/geth/chaindata/ancient/chain readonly=false
INFO [03-26|04:28:57.641] Initialising Ethereum protocol           network=123,454,321 dbversion=8
INFO [03-26|04:28:57.641]
INFO [03-26|04:28:57.641] --------------------------------------------------------------------------------------------------------------------------------
INFO [03-26|04:28:57.641] Chain ID:  12345 (unknown)
INFO [03-26|04:28:57.641] Consensus: Clique (proof-of-authority)
```

15. Now do same for node2

    geth --datadir node2 --port 30307 --bootnodes enode://3f690f88c00ce563d9dd609b0674ad97df9fc6b77673bd86917f51333118e81f54e75919746a331d55af7f6edba870a5ef5ed12b60af554ccb922bc77ab41f27@127.0.0.1:0?discport=30305 --networkid 123454321 --unlock 0x6d04eA66725A40F5669E2d6f4f77eb4Ecf756a3f --password node2/password.txt --authrpc.port 8552

```
deilsy@delsy:~/private_ethereum_setup$ geth --datadir node2 --port 30307 --b
ootnodes enode://3f690f88c00ce563d9dd609b0674ad97df9fc6b77673bd86917f5133311
8e81f54e75919746a331d55af7f6edba870a5ef5ed12b60af554ccb922bc77ab41f27@127.0.
0.1:0?discport=30305 --networkid 123454321 --unlock 0x6d04eA66725A40F5669E2d
6f4f77eb4Ecf756a3f --password node2/password.txt --authrpc.port 8552
INFO [03-26|04:35:17.744] Maximum peer count                       ETH=50 LE
S=0 total=50
INFO [03-26|04:35:17.745] Smartcard socket not found, disabling    err="stat
 /run/pcscd/pcscd.comm: no such file or directory"
INFO [03-26|04:35:17.748] Set global gas cap                       cap=50,00
0,000
INFO [03-26|04:35:17.748] Initializing the KZG library             backend=g
okzg
INFO [03-26|04:35:17.795] Allocated trie memory caches             clean=154
.00MiB dirty=256.00MiB
INFO [03-26|04:35:17.795] Using pebble as the backing database
INFO [03-26|04:35:17.795] Allocated cache and file handles         database=
/home/deilsy/private_ethereum_setup/node2/geth/chaindata cache=512.00MiB han
dles=524,288
INFO [03-26|04:35:17.928] Opened ancient database                  database=
/home/deilsy/private_ethereum_setup/node2/geth/chaindata/ancient/chain reado
nly=false
INFO [03-26|04:35:17.929] Initialising Ethereum protocol           network=1
23,454,321 dbversion=<nil>
INFO [03-26|04:35:17.930]
INFO [03-26|04:35:17.930] ---------------------------------------------------
----------------------------------------------------------------------------
--------------------------
INFO [03-26|04:35:17.930] Chain ID:  12345 (unknown)
INFO [03-26|04:35:17.930] Consensus: Clique (proof-of-authority)
INFO [03-26|04:35:17.930]
INFO [03-26|04:35:17.930] Pre-Merge hard forks (block based):
INFO [03-26|04:35:17.930]  - Homestead:                   #0        (https:/
/github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-up
grades/homestead.md)
```

16. Now we will interact with the blockchain using javascript console geth attach node1/geth.ipc

```
deilsy@delsy:~/private_ethereum_setup$ geth attach node1/geth.ipc
Welcome to the Geth JavaScript console!

instance: Geth/v1.12.0-stable-e501b3b0/linux-amd64/go1.20.3
coinbase: 0x563a5a670260417b163ad1696fcbfa6849906d06
at block: 105 (Thu Mar 26 2026 04:37:38 GMT+0000 (UTC))
 datadir: /home/deilsy/private_ethereum_setup/node1
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.
0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
>
```

17. Interacting with the blockchain

```
> net.peerCount
1
> eth.accounts
["0x563a5a670260417b163ad1696fcbfa6849906d06"]
> eth.accounts[0]
"0x563a5a670260417b163ad1696fcbfa6849906d06"
> eth.blockNumber
302
> eth.getBalance(eth.accounts[0])
1e+23
> web3.fromWei(eth.getBalance("0x563A5a670260417B163AD1696fCBfA6849906d06"),
 "ether")
100000
```

```
> eth.getBlock(eth.blockNumber)
{
  baseFeePerGas: 7,
  difficulty: 2,
  extraData: "0xd883010c008467657468886676f312e32302e33856c696e75780000000000000
000040f57c8f0c1bbe3fa04b217bfdc354aa1e411ae790f2aa93ecf3b9f49ccbfd224b4c30f2
d379e2f80d9626b9aa24660a7e3f8e6276b34aba4825c2956e45a62200",
  gasLimit: 586348211,
  gasUsed: 0,
  hash: "0x5cd8ce4ebaace9c9674c46f1346b87b916bf6e15113dd3270634ac6b5cb0e96c"
,
  logsBloom: "0x0000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000",
  miner: "0x0000000000000000000000000000000000000000",
  mixHash: "0x00000000000000000000000000000000000000000000000000000000000000
00",
  nonce: "0x0000000000000000",
  number: 318,
  parentHash: "0x6a5fac551749b1a9ee9a3e7c8b0db9c6b3f4ffd7175e8df285c1bb74b70
d64cd",
  receiptsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e
363b421",
  sha3Uncles: "0x1dcc4de8dec75d7aab85b567b6ccd41ad312451b948a7413f0a142fd40d
49347",
  size: 610,
  stateRoot: "0x121c3db0cbfb3018fc585199bda672875d3caa3a3e83f480387ce86ac9d2
7e34",
  timestamp: 1774500923,
  totalDifficulty: 637,
  transactions: [],
  transactionsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622
fb5e363b421",
  uncles: []
```

18. Sending transactions between accounts

```
> eth.sendTransaction({from:"0x563A5a670260417B163AD1696fCBfA6849906d06", to
:"0x6d04eA66725A40F5669E2d6f4f77eb4Ecf756a3f", value: web3.toWei(10, "ether"
), gas:30000})

"0x3cb6cc9cd65c927b0a0af8351e757ef8fdf76ea00d66b8863b3c9fdfc3657508"
> web3.fromWei(eth.getBalance("0x563A5a670260417B163AD1696fCBfA6849906d06"),
 "ether")

99989.999999999999853
> web3.fromWei(eth.getBalance("0x6d04eA66725A40F5669E2d6f4f77eb4Ecf756a3f"),
 "ether")
120010
```

19. Analyzing transactions

```
> web3.fromWei(eth.getBalance("0x6d04eA66725A40F5669E2d6f4f77eb4Ecf756a3f"),
 "ether")
120010
> web3.eth.getTransaction("0x3cb6cc9cd65c927b0a0af8351e757ef8fdf76ea00d66b88
63b3c9fdfc3657508")
{
  accessList: [],
  blockHash: "0xe772e9613d2f39454f41076dd9e740948bac2e0ecfa254702b4f474fbecc
a438",
  blockNumber: 338,
  chainId: "0x3039",
  from: "0x563a5a670260417b163ad1696fcbfa6849906d06",
  gas: 30000,
  gasPrice: 1000000007,
  hash: "0x3cb6cc9cd65c927b0a0af8351e757ef8fdf76ea00d66b8863b3c9fdfc3657508"
,
  input: "0x",
  maxFeePerGas: 1000000014,
  maxPriorityFeePerGas: 1000000000,
  nonce: 0,
  r: "0x84be672e734bdae2f6d262abda2a08687028a86e035d8c73c8c80662fd6c64fe",
  s: "0x196df54d8f571bde60289c94926793a29f05947203c25418295a1cd80c7f9c6a",
  to: "0x6d04ea66725a40f5669e2d6f4f77eb4ecf756a3f",
  transactionIndex: 0,
  type: "0x2",
  v: "0x0",
  value: 10000000000000000000
}
>
```

Conclusion - Private blockchain was created using geth, the nodes in the blockchain were configured and we interacted with the blockchain by making a transaction of 10 ethers.